

Pilot-Scale Assessment of Electrochemical Ozone Oxidation for Textile Wastewater Treatment: Process Performance and Ecotoxicological Evaluation

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01 INTRODUCTION

● **Problem:** Textile wastewater rich in POPs, dyes, heavy metals, surfactants, acids, and alkalis.

● **Limitation:** Conventional treatment is inadequate; chlorination forms DBPs and results in salt/sludge waste accumulation.

● **Solution:** **ECOOP (Electrochemical Ozone Oxidation)** integrates ozonation and electrolysis for efficient contaminant removal and reduced ecotoxicological impact.

02 OBJECTIVES

✓ Evaluate the performance of the pilot-scale Electrochemical Ozone Oxidation Process (ECOOP) for the treatment of synthetic and real textile wastewater streams.

✓ Assess the removal of organic pollutants, color, and microplastics

✓ Assess the reduction in ecotoxicity of textile wastewater following ECOOP treatment

03 METHODOLOGY

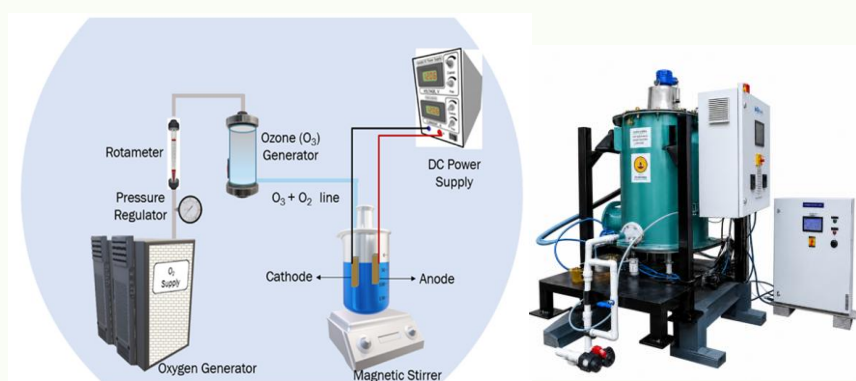


Figure 1: Process Schematic & Pilot Reactor developed

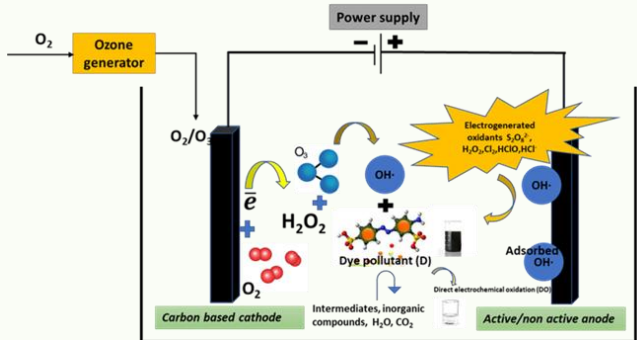


Figure 2: Reaction Mechanism

Table 1: Operating conditions

Parameter	Value
Current density	3.5 mA/cm ² , 75 mA/cm ²
Ozone Injection rate	1gph, 50 gph
Cathode	Stainless steel
Anode	MMO, BDD
Reactor Volumes	500 mL, 350 L

Analytical Techniques used:

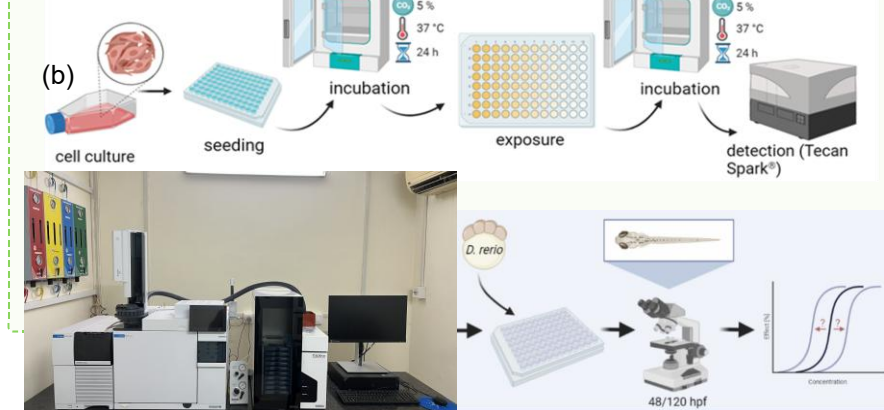


Figure 3: Analytical Techniques used for (a) process evaluation and (b) ecotoxicological studies

04 RESULTS

Pilot-scale ECOOP effectively treated dye bath wastewater, achieving superior removal of color, organic pollutants, microplastics, and toxic contaminants with minimal secondary pollution.

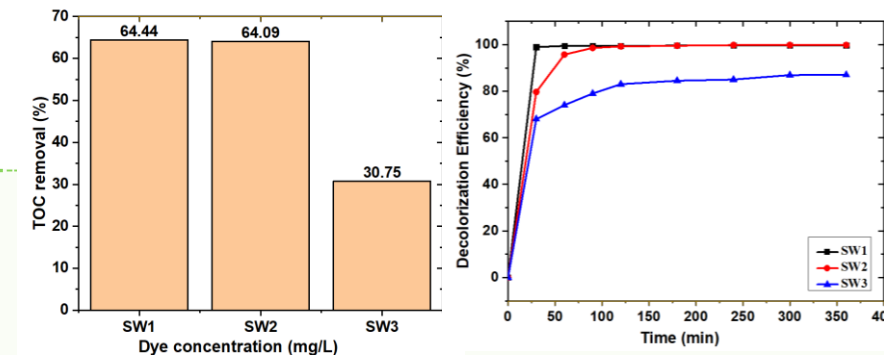


Figure 4: Synthetic wastewater treatment (a) Organics & (b) Color removal

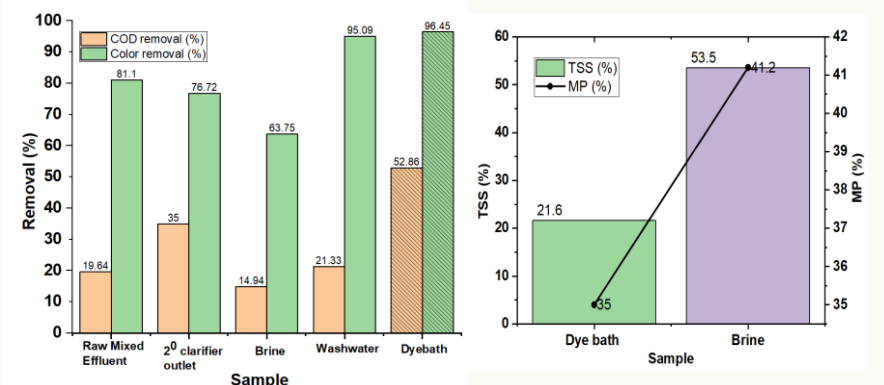


Figure 5: Real effluent treatment (a) Organics & Color removal (b) microplastics and TSS reduction

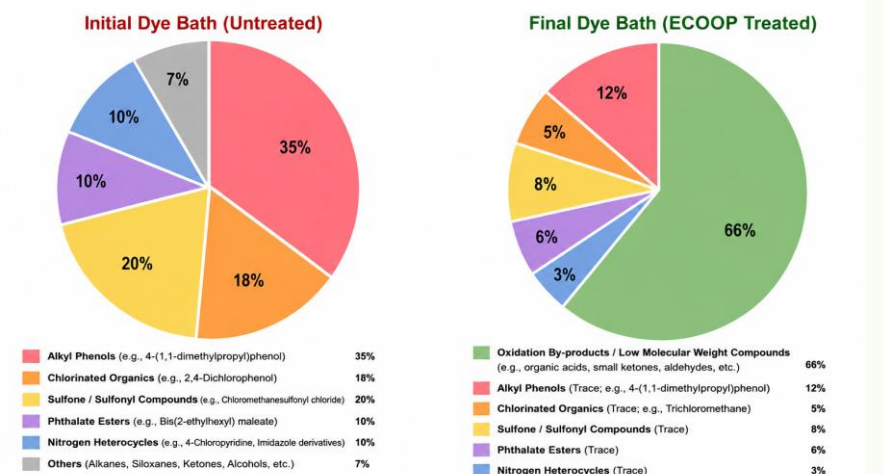


Figure 6: GCMS Results



Table 2: Cost comparison details

	Current	Proposed
Treatment	Mixed effluent	Segregated streams
Cost (10 KLD/day)	₹ 1,213.70	₹ 1,297.60
Difference	—	6.90%

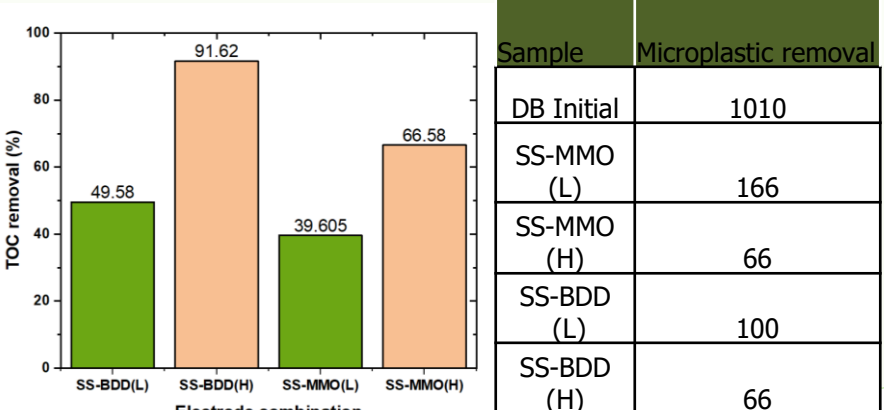


Figure 7: Comparison of MMO with BDD when used as anodes at lab scale.

05 DISCUSSION

• Based on the process performance, a stream segregation strategy is proposed in which high-strength dye bath wastewater is treated separately from low-strength wash water, enabling targeted and efficient treatment.

• Observed reduction in ecotoxicological effects demonstrates its potential to reduce the overall environmental footprint.

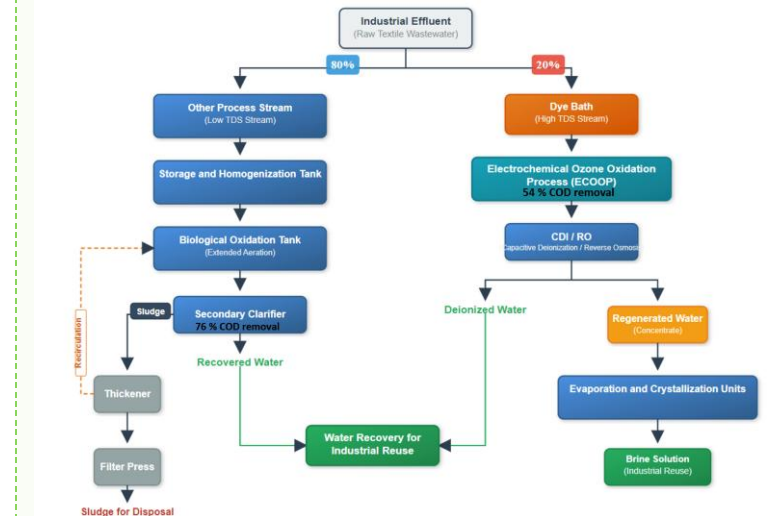


Figure 8: Proposed Treatment strategy

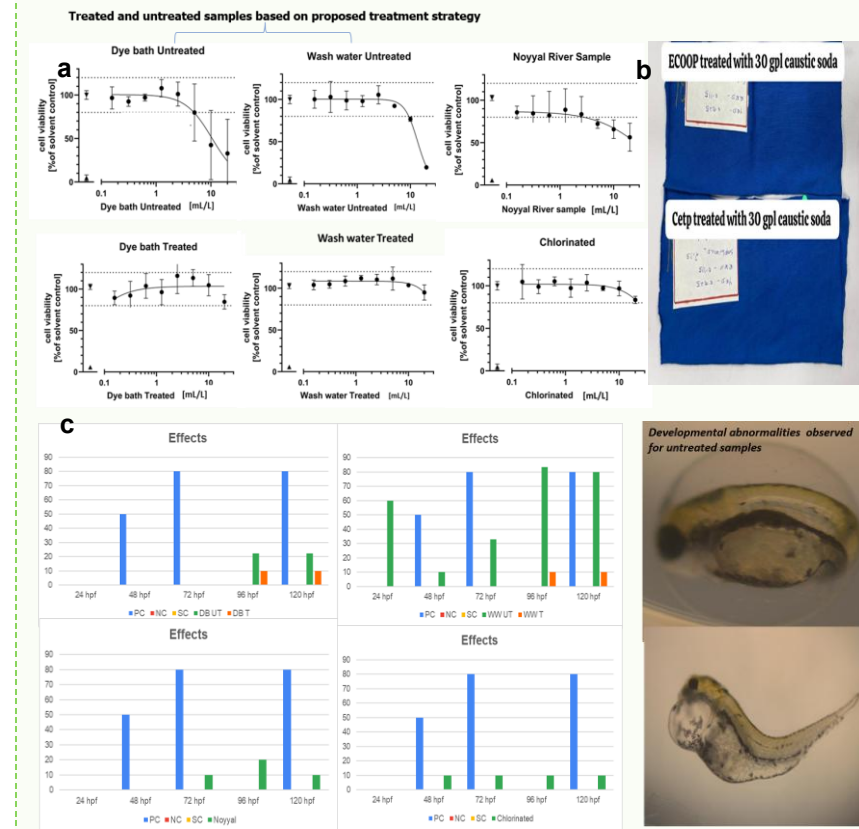


Figure 9: (a) Neutral Red Retention Assay (b) Visual color Assessment (c) FET

06 CONCLUSIONS

Efficient removal of color, organic pollutants, microplastics, achieved with minimal secondary pollution.

• ECOOP combined with stream segregation is a promising approach for sustainable textile wastewater treatment

Pilot-scale ECOOP demonstrates the potential of advanced oxidation for sustainable textile wastewater treatment with improved environmental safety

—KEY FINDING

07 FUTURE WORK

- Testing BDD as anode in the pilot reactor
- Modifications at cathode to improve mineralization performance
- Industrial validation
- Techno-economic and life-cycle assessment for full-scale implementation.

08 REFERENCES

1. Bakheet, B., Yuan, S., Li, Z., Wang, H., Zuo, J., Komarneni, S., & Wang, Y. (2013). Electro-peroxone treatment of Orange II dye wastewater. Water Research, 47(16), 6234–6243.
2. Gossen, M., Govindarajan, D., John, A. A., Hussain, S., Padigur, M., Linnartz, C., Mohseni, M., Stüwe, L., Urban, V., Crawford, S., Schiwy, S., Wessling, M., Nambi, I. M., & Hollert, H. (2023). ElectroH2O: Development and evaluation of a novel treatment technology for high-brine industrial wastewater. Science of the Total Environment, 883(April), 1–9.

09 ACKNOWLEDGEMENTS

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